

1 Trustworthy uncertainty quantification for quantitative
2 and diffusion body MRI, and its decision-use in
3 MR-guided adaptive radiotherapy: a field review

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5 **Abstract**

In MR-guided adaptive radiotherapy, a quantitative or diffusion MRI biomarker is increasingly offered not as an image but as a *number* that can steer dose — a dose-painting prescription, or a response-adaptive change to the plan. Acting on that number means acting on the *uncertainty* attached to it, and the question this review surveys is whether that uncertainty is trustworthy enough to drive a dose change. We read the field as the path a biomarker error bar must travel to reach the dose decision: *trust* (is the bar honest — calibrated and covering?), *value* (does a trusted bar change the decision?), and *action* (what happens when it is deployed to drive dose on an MR-linac?), over a foundations layer (what is measured, and with what estimator) and beneath a map of open problems. We find that trust is conditional: broadly defensible for the diffusion coefficient D and ADC, but fragile for the perfusion parameters f and D^* , whose measured test–retest reliability is poor and for which deep-learning uncertainty quantification typically reports only marginal, aggregate calibration. The decision-analytic apparatus to price a calibrated interval — net benefit, decision calibration, value of information — is mature but rarely connected to quantitative-MRI uncertainty; and the propagation of *biomarker* uncertainty into the adaptive dose decision remains the immature link. We close with four open problems for biomarker-driven adaptive radiotherapy, sited where trust,

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value, and action meet. This review reports no new measurement; its contribution is synthesis, taxonomy, and gap identification.

Keywords: MR-guided adaptive radiotherapy, dose painting, quantitative MRI, intravoxel incoherent motion, uncertainty quantification, value of information

1. The dose decision the biomarker must serve

Adaptive MR-guided radiotherapy increasingly proposes to steer dose from what a quantitative or diffusion MRI sequence *measures*, not only from where the anatomy sits. On an MR-linac the plan can be re-optimised between or within fractions, and an emerging paradigm would let a functional or quantitative biomarker — an apparent diffusion coefficient (ADC), an intravoxel-incoherent-motion (IVIM) parameter, a perfusion measure — raise or lower dose to a sub-volume according to estimated biology [1, 2]. The biological premise is **dose painting**: prescribing a non-uniform dose from a target volume derived from functional or molecular images [3–5]. What such a prescription actually consumes is not the biomarker but a *number with an error bar*; and the unsolved link is whether that error bar is honest and stable enough to justify changing a dose.

This review surveys that link. We read the literature as the path a biomarker error bar must travel before it can reach a dose decision:

1. **Trust** — is the error bar honest (calibrated, covering)?
2. **Value of information** — given a trusted bar, does it change the decision?
3. **Action** — when deployed to drive dose on an MR-linac, where does it hold and where does it break?

over a **foundations** layer (what is measured, and with what estimator) and beneath a **gap map** of the field’s open problems. Because the dose decision is the organizing concern, we foreground the action setting (Sec. 2) first, then walk foundations (Sec. 3), trust (Sec. 4), and value (Sec. 5) as the question of whether the quantitative-MRI side can supply what that decision demands. The survey

Table 1: The survey axis. The field is classified by the question each work answers about *somebody’s* uncertainty estimate, foregrounding the action setting — the dose decision — that the other axes serve; a cross-cutting gap map (Sec. 6) sits over the axes at the seams between them.

axis	what the bucket asks
Action	When deployed to drive dose on an MR-linac, where does a biomarker’s error bar hold and break? (Sec. 2)
Foundations	What is the estimand, and what estimator (with what uncertainty) produced it? (Sec. 3)
Trust	Is the error bar honest — calibrated and covering? (Sec. 4)
Value of information	Given a trusted bar, does it change an oncologic dose decision? (Sec. 5)
Gap map	Where, at the seams between the axes, do the field’s UQ-trust questions remain open for adaptive RT? (Sec. 6)

32 covers uncertainty quantification (UQ) for quantitative/diffusion body MRI and
33 its decision-use in MR-guided adaptive RT; it excludes non-quantitative MRI,
34 neuro-only work, generic computer-vision UQ, and non-MR-guided RT (the
35 scope boundary is recorded in the repository’s `ASSUMPTIONS.md`). Table 1 lays
36 out the survey axis and the thematic buckets it induces.

37 2. The action setting: biomarker-driven adaptive radiotherapy

38 The action setting where a quantitative-MRI biomarker most directly drives
39 dose is the MR-Linac. Integrating diagnostic-quality MRI with a linear accelera-
40 tor [6, 7] matured into a clinically feasible 1.5 T MRI-guided system [8, 9] and an
41 emerging paradigm for adaptive radiation oncology [1, 2]. The biological premise
42 is **dose painting** — prescribing a non-uniform dose from a biological target vol-
43 ume derived from functional/molecular images [3–5] — with DWI/ADC a lead-

44 ing candidate response biomarker [10] and quantitative MRI proposed to update
 45 the plan to observed tumour response [1, 11]. Clinical translation is real but un-
 46 even: a randomised focal boost to an MRI-defined intraprostatic lesion improved
 47 biochemical control in the FLAME trial [12] and dose-painting-by-numbers has
 48 been delivered in early-phase studies [13], yet a phase III ^{18}F -FDG-PET dose-
 49 redistribution trial found no overall tumour-control benefit [14] — so steering
 50 dose from an image map is supported in principle but not settled in outcome.
 51 Diffusion biomarkers are now being tested on the MR-linac itself, e.g. longitudi-
 52 nal per-fraction DWI during pancreatic stereotactic RT as a prognostic marker
 53 [15].

54 Two cautions make the dose decision conditional on the biomarker’s uncer-
 55 tainty, not just its value. First, **uncertainty must be carried into the**
 56 **optimisation:** robust/probabilistic planning incorporates motion and uncer-
 57 tainty directly rather than relying on PTV margins [16], which is the natural
 58 entry point for a *biomarker* error bar but has so far ingested mainly geometric
 59 and motion uncertainty. Second, **technical validation gates clinical use:**
 60 imaging biomarkers on an MR-linac require repeatability/reproducibility vali-
 61 dation before driving decisions, because the integrated scanner differs from a
 62 diagnostic system [17]. That validation programme is now underway: quantita-
 63 tive $T_1/T_2/\text{ADC}$ mapping is feasible and accurate on the 1.5 T MR-linac [18],
 64 the Elekta Unity consortium has issued ADC-acquisition recommendations for
 65 the integrated system [19], and a multi-institutional registry pools the clinical
 66 and technical data needed to evaluate it [20]. **Takeaway:** the field is converging
 67 on biomarker-driven adaptive dose, and the two cautions define the demand it
 68 places on the quantitative-MRI side — a calibrated, technically validated error
 69 bar — yet the propagation of *biomarker uncertainty* into the dose decision is
 70 the immature link. The rest of the review asks whether that side can meet the
 71 demand: what is measured (Sec. 3), whether its error bar is honest (Sec. 4), and
 72 whether a trusted bar would change the decision (Sec. 5).

73 3. Foundations: estimands and estimators

74 Before honesty can be assessed, the estimand and estimator must be fixed,
75 because the error bar a method reports is a property of the estimator, not
76 of the biomarker alone. The IVIM model was introduced to separate molec-
77 ular diffusion from capillary-network microcirculation within a voxel [21, 22],
78 treating randomly oriented capillary flow as a pseudo-diffusion process [23]; dy-
79 namic contrast-enhanced (DCE) MRI quantification rests on the compartmental
80 transfer-constant model [24] and its standardised quantities K^{trans} , v_e , k_{ep} [25].

81 The estimator is not innocent. Cramér–Rao/Fisher analysis shows the
82 achievable precision of IVIM D and f is strongly SNR-dependent [26], and the
83 same bounds can be inverted to design SNR-efficient acquisitions [27]. Empir-
84 ically, IVIM parameter estimates differ *significantly* across fitting algorithms
85 in the upper abdomen, with Bayesian fitting giving the lowest inter-reader
86 and inter-subject variability [28]; simulation [29] and pancreatic-cancer data
87 [30] corroborate the Bayesian advantage for D and f . A parallel line replaces
88 hand-built fits with learned ones — neural-network IVIM/kurtosis mapping
89 [31], supervised and unsupervised physics-informed deep fitting [32, 33], and
90 self-supervised forward-model estimation [34] — with recent work decomposing
91 predictive uncertainty into aleatoric and epistemic components [35] or returning
92 full posteriors that expose model degeneracy [36]. DCE shares the trajectory,
93 e.g. Bayesian spatial tracer-kinetic estimation [37]. **Takeaway:** the uncertainty
94 a method reports is inseparable from the estimator that produced it, so the
95 error bar an adaptive-RT decision consumes must be judged per-estimator, not
96 per-biomarker.

97 4. Trust: is the error bar honest?

98 4.1. The general machinery of trust

99 The UQ community supplies the toolkit the imaging literature draws on.
100 Distribution-free **conformal prediction** wraps any black-box predictor in a
101 set/interval valid at a user-chosen error rate [38, 39]. **Calibration** methods

102 recalibrate regression intervals post-hoc [40] and correct the systematic over-
 103 confidence of modern deep networks [41]. Practical deep-UQ families — deep
 104 ensembles [42] and MC-dropout as approximate Bayesian inference [43] — give
 105 usable uncertainty, but their quality **degrades under dataset shift** [44] —
 106 the operative warning for an MR-linac, whose integrated scanner is itself a shift
 107 from the diagnostic systems on which such methods are trained.

108 4.2. *The metrology of trust in quantitative imaging*

109 Imaging has its own standards for honest reliability. The QIBA framework
 110 anchors technical performance on three metrology areas — linearity/bias, re-
 111 peatability, reproducibility [45] — with companion guidance on comparing al-
 112 gorithms [46] and on consistent terminology [47]. Reliability reporting leans on
 113 the intraclass correlation coefficient [48] and on limits-of-agreement rather than
 114 correlation for method comparison [49].

115 4.3. *Where trust is empirically tested in body MRI*

116 When these standards are applied to body diffusion/perfusion MRI, the
 117 error bars are often **not** honest — and the dishonesty is parameter- and cohort-
 118 dependent. ADC is a comparatively reliable biomarker but its in-vivo repro-
 119 ducibility is markedly worse than phantom reproducibility [50], and the body-
 120 DWI consensus already urged baseline reproducibility studies as part of study
 121 design [51, 52]. The perfusion parameters are the weak point: short-term test-
 122 retest repeatability coefficients for IVIM f and D^* reach $\sim 126\%$ and $\sim 197\%$ in
 123 rectal cancer [53], D^* reproducibility is only moderate ($\text{ICC} \approx 0.55$, $\text{CV} \approx 20\%$)
 124 [54], and multi-scanner liver IVIM within-subject CVs run from $\sim 5\%$ for D
 125 up to $\sim 14\%$ for perfusion-fraction components [55]. On the MR-linac itself,
 126 even ADC — the reliable end of the spectrum — carries a sizeable test-retest
 127 repeatability coefficient: $\sim 17\%$ in rectal cancer [56] and $\sim 13\text{--}31\%$ across head-
 128 and-neck targets [57], setting an explicit floor below which an observed change
 129 is measurement noise, not response. Cross-modal anchoring does not rescue
 130 D^* : its correlation with DCE K^{trans} is weak in one rectal cohort ($r = 0.389$)

131 [58] and non-significant in another [54]. **Takeaway:** trust in body-MRI UQ
 132 is conditional — broadly defensible for D/ADC , fragile for f/D^* — yet most
 133 deep-learning UQ reports aggregate or marginal calibration [35], leaving the
 134 conditional failure region (exactly the regime an adaptive dose change would
 135 rely on) under-characterised.

136 5. Value of information: does a trusted bar change the dose decision?

137 A calibrated interval earns its cost only if it moves an action — here, a
 138 dose decision. The decision-analytic literature supplies the apparatus: **dec-**
 139 **sion curve analysis / net benefit** evaluates a model by clinical net benefit
 140 across the decision threshold [59–61], where the threshold encodes the clini-
 141 cian’s relative weighting of a missed disease against an unnecessary interven-
 142 tion; performance frameworks argue such decision-analytic measures should
 143 be reported whenever a model supports decisions [62]. From the inference
 144 side, **decision/loss-calibrated** methods tie the posterior approximation to the
 145 downstream decision — decision calibration [63] and loss-calibrated Bayesian
 146 inference [64]. From health economics, **value-of-information** analysis (EVPI)
 147 prices a measurement by combining the probability it changes a decision with
 148 the benefit of that change [65], under the decision-theoretic stance that choices
 149 follow expected net benefit rather than statistical significance [66]. **Takeaway:**
 150 the machinery to ask “does this body-MRI interval change an oncologic dose
 151 decision?” exists and is mature — but is rarely connected to quantitative-MRI
 152 uncertainty, which is precisely what an adaptive-RT plan change would need.

153 6. Gap map: open problems for biomarker-driven adaptive RT

154 The field’s unresolved UQ-trust questions cluster at the **seams between**
 155 **the axes** — and, read for adaptive RT specifically, each is a problem that must
 156 be solved before a biomarker error bar can be trusted to steer dose.

- 157 • **G1 — conditional honesty (trust seam).** Repeatability evidence
 158 shows f/D^* are far less reliable than D/ADC [53–55], and even ADC car-

159 ries a non-trivial on-device repeatability floor [56, 57], yet deep-learning
 160 IVIM-UQ typically reports marginal/aggregate calibration [35]. *Open:* cal-
 161 ibration conditioned on the regime where it fails, and conformal/coverage
 162 guarantees that respect it [39, 44] — the regime an adaptive dose change
 163 would actually rely on.

164 • **G2 — from calibration to dose decision (trust→value seam).** The
 165 decision-analytic toolkit [59, 63] is rarely applied to quantitative-MRI
 166 uncertainty. *Open:* when does a calibrated body-MRI interval actually
 167 change an oncologic dose decision versus merely being reported?

168 • **G3 — uncertainty into dose (value→action seam).** Adaptive-RT
 169 reviews call for technically validated biomarkers to steer dose [1, 17], and
 170 robust planning can ingest uncertainty [16]. *Open:* end-to-end propa-
 171 gation of *biomarker* uncertainty (not just geometric/motion uncertainty)
 172 into the adaptive dose decision, and the identifiability walls that cap it.

173 • **G4 — estimator-dependence of trust (cross-cutting).** Reported
 174 uncertainty depends on the estimator [28, 32]. *Open:* standards that make
 175 UQ comparable across classical and learned estimators, extending QIBA
 176 metrology [45, 47] to the uncertainty itself, not only the point estimate —
 177 a prerequisite for trusting any of them on an MR-linac.

178 7. Discussion and limitations

179 This review establishes three things for the biomarker-to-dose problem. First,
 180 trust in body-MRI uncertainty is *conditional*: the diffusion coefficient D and
 181 ADC carry broadly defensible error bars, whereas the perfusion parameters f
 182 and D^* do not, and most deep-learning UQ reports only marginal, aggregate
 183 calibration that does not characterise where the failure concentrates (Sec. 4).
 184 Second, the decision-analytic apparatus needed to price a calibrated interval
 185 against a dose decision — net benefit, decision calibration, value of information
 186 — is mature, but is rarely connected to quantitative-MRI uncertainty (Sec. 5).

187 Third, the propagation of *biomarker* uncertainty into the adaptive dose deci-
188 sion, as opposed to geometric or motion uncertainty, remains the immature link
189 (Sec. 2). The four open problems (Sec. 6) sit at exactly these seams.

190 The review deliberately draws a scope boundary. It covers uncertainty quan-
191 tification for quantitative and diffusion body MRI and its decision-use in MR-
192 guided adaptive radiotherapy; it excludes non-quantitative MRI, neuro-only
193 work, generic computer-vision UQ with no quantitative-MRI tie, and radio-
194 therapy that is not MR-guided. That boundary keeps the survey on the dose-
195 decision link and is recorded in the repository’s `ASSUMPTIONS.md`.

196 The principal limitation is methodological: this is a single-author *narrative*
197 synthesis, not a systematic review. It follows no pre-registered search protocol,
198 and the selection of works — though chosen to span the estimation, calibra-
199 tion, decision-analytic, and MR-guided-RT literatures the problem touches —
200 is expert-curated and therefore subject to selection bias. A systematic review
201 with an explicit search-and-inclusion protocol, ideally with more than one re-
202 viewer, would be the natural next step to harden the coverage claims made
203 here. Every empirical statement is nonetheless attributed to a verified external
204 source.

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214 editing.

215 **Data availability**

216 This review reports no new data. The verified citation base and the script
217 that checks that every reference resolves are openly available in the project
218 repository.

219 **Declaration of generative AI and AI-assisted technologies in the** 220 **manuscript preparation process**

221 During the preparation of this work the author used Claude (Anthropic),
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